

Probability PhD Exam, May 1998

1. Let X and Y be independent and uniform on $[0, 1]$. Find:

- a) the distribution of $X + Y$.
- b) the conditional density of X given $X + Y = z$.

2. Let $X_0, X_1, \dots, X_n, \dots$ be i.i.d. with mean m . Let N be Poisson with mean λ , independent of the X 's. Define

$$Y = X_0 + \dots + X_N.$$

Prove that Y is integrable. Find $E(Y)$.

3. State precisely (all for i.i.d.):

- a) the strong law of large numbers and prove it.
- b) the central limit theorem.
- c) the law of the iterated logarithm.

4. Define a martingale relative to a filtration $\{\mathcal{F}_n\}$, $n = 0, 1, 2, \dots$. Define a stopping time relative to this filtration. State the optional sampling theorem. State and prove the martingale convergence theorem.

5. What is an infinitely divisible distribution on $\mathbb{R}^1$. Prove that a non-trivial infinitely divisible distribution cannot have compact support.

6. Define a standard Brownian Motion W starting at 0. Define new processes W_1, W_2, W_3 by:

$$W_1(t) = cW\left(\frac{t}{c^2}\right), \quad t \geq 0, \quad c \text{ real, } c \neq 0,$$

$$W_2(t) = tW\left(\frac{1}{t}\right), \quad t > 0, \quad W_2(0) = 0,$$

$$W_3(t) = \begin{cases} W(1) - W(1-t), & 0 \leq t \leq 1, \\ W(t), & \text{otherwise.} \end{cases}$$

Show that W_1, W_2, W_3 are the standard Brownian motions.

7. Let $\varphi(t) = \int e^{itx} \mu(dx)$ be a characteristic function where μ is a probability measure on $\mathbb{R}^1$.

- a) Suppose $|\varphi(t)| = 1$ for some $t \neq 0$. Then prove that unless $|\varphi(t)| \equiv 1$, there is a smallest $t_0 > 0$ and a d such that μ is concentrated on the set $\left\{d + \frac{2\pi j}{t_0}\right\}$, $j = 0, \pm 1, \pm 2, \dots$
- b) Using a) prove that $|\cos t|$ is not a characteristic function. Hint: $\cos t$ and $\cos^2 t$ are characteristic functions.